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RESEARCH ARTICLE

Leveraging Fusion Methods of Human Pose and Motion Dynamics for Accurate Violence Detection in Video Surveillance

SAI THU YA AUNG[1], WORAPAN KUSAKUNNIRAN[1], (Member, IEEE), AND YEW KWANG HOOI[2]

[1]Faculty of Information and Communication Technology, Mahidol University, Phutthamonthon, Nakhon Pathom 73170, Thailand
[2]Department of Computer and Information Sciences, Universiti Teknologi PETRONAS, Seri Iskandar, Perak 32610, Malaysia

Corresponding author: Worapan Kusakunniran (worapan.kun@mahidol.edu)

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ABSTRACT The growing demand for security is leading to an increase in the installation of surveillance cameras and there is an urgent need for automated systems. Despite large number number of surveillance cameras are installed to detect and prevent violent activities, the need for monitoring and analyzing the footage in real time is a challenge. To accurately detect violence, it is essential to consider human interactions and minimize noise in surveillance footage. This study explores violence detection using a dual-stream deep neural network with human skeletons and motion changes as inputs extracted from surveillance videos. Furthermore, a weighted fusion strategy is employed, which integrates fusion operation functions, activation functions, and weighted outputs to prioritize the most relevant features from multiple inputs for effective violence detection. Our proposed framework is both effective and accurate, achieving accuracy of 95.05%. The use of skeletal data with a black background significantly improved performance compared to traditional RGB frames, while the incorporation of motion information and fusion strategy improved the performance of violence detection.

INDEX TERMS Deep neural network, human pose, motion dynamics, violence detection.

I. INTRODUCTION

The number of surveillance cameras installed worldwide is quickly surpassing the capabilities of human-operated monitoring systems. AI-based violence monitoring systems present promising and efficient alternatives to conventional human-centric surveillance systems. Unlike human-based systems, AI-assisted surveillance solutions offer reliability, predictable downtime, and seamless scalability.

Violent actions can be deduced from specific postures, gestures, movements, and interactions between individuals [1], [2]. From the postures and movements of an individual certain violent actions can be detected. Another key indicator is motion information, which captures the intensity and dynamics of human body movements and is essential for recognizing rapid and forceful actions, such as pushing and hitting. In this study, human poses extracted as skeleton information and motion were applied for the detection of violent actions.

Training deep learning models to detect violence in surveillance videos depends on the quality of the videos. Violence detection in surveillance videos is hindered by several challenges, such as background noise can interfere with the clarity of the video and making the model difficult to detect humans. Other concerns involve variations in lighting and resolution which can deter human detection for violence detection.

To address the challenges of violence detection in surveillance videos, a comprehensive framework and model architecture are proposed. The contributions of this study are as follows:

1) Skeletal-centric noise reduction strategy is implemented by isolating skeletal information through pose estimation and rendering skeletons over black background to remove irrelevant background features.

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2) Various approaches using weighted fusion are explored, applying activation functions to assign weights to input branches and identifying the most effective configurations.
3) A combined fusion strategy is proposed that employs a weighted fusion approach, utilizing different activation functions and applying various values of additional weights by multiplying them to the branches of the models before element-wise addition for branch fusion.

II. RELATED WORK

Human skeleton extraction is typically facilitated by pose estimation techniques, where deep learning models are trained to recognize and locate key body points, such as joints and limbs. The resulting ''skeleton'' model represents the human body structure in a simplified form, enabling more efficient analysis of movement, posture, and interactions with the environment. In surveillance applications, human skeleton extraction is employed for purposes such as security monitoring, behavioral analysis, and crowd control [3], [4].

A dual-stream model for human action recognition was proposed by Xu et al. [5], which integrated scene images and skeleton data. Eighteen-joint skeletons were extracted using OpenPose and represented as a spatio-temporal graph, while motion features were captured through graph convolution. Scene frames were processed using a sparse sampling and video-level consensus strategy. By fusing both streams, recognition performance was improved. Similarly, the Multimodal SpatioTemporal Skeletal Kinematic Gait Feature Fusion (MSTSK-GFF) framework was presented by Amsaprabhaa et al. [6] for fall detection using video data. Moreover, a deep learning architecture for detecting violent crimes in surveillance videos was proposed by Garcia et al. [7], utilizing human pose estimation and change detection. Human skeletons were extracted through pose estimation, and temporal changes were incorporated. Fusion technique combined these inputs, while a ConvLSTM enabled efficient temporal aggregation, reducing model parameters fivefold. The model achieved state-of-the-art results with real-time inference and was evaluated on benchmark datasets.

Motion information extraction from video data involves analyzing movement within frames for behavior detection, and video retrieval. Key techniques include frame differencing [8], which detects motion by comparing consecutive frames to estimate global changes; optical flow [9], which analyzes pixel movement between frames for detailed motion tracking; and block matching [8], which segments frames into blocks to estimate local displacement and motion patterns. These methods play a crucial role in motion analysis, and contribute to advancements in surveillance, activity recognition, and video processing.

Multi-stream deep learning networks are employed to extract and fuse deep features from multiple data representations. This involves using multiple input channels to explore interactions among these representations, leading to improved performance in action recognition. Islam et al. [10] proposed a fusion method using concatenating, addition and multiplication to combining the information in violence detection from video inputs. Their model involve several types of ConvLSTM to capture spatial and temporal information with few parameters. Diba et al. [11] presented a two-stream approach to capture motion and appearance using 3D CNN for video classification. They used original video frames and optical flow as inputs. Their 3D CNN model employed appearance stream to capture features from original video frames and motion stream for optical flow estimation, and combined features from both streams.

III. PROPOSED FRAMEWORK

Our proposed framework consists of two steps: the first step involves data preprocessing for video inputs and the second step involves model training with preprocessed inputs.

A. PREPROCESSING FOR SKELETAL AND MOTION INFORMATION EXTRACTION

In the initial phase of preprocessing, each frame of the video is processed sequentially. For every frame, Gamma Correction is applied and pose estimation is calculated using YOLO-NAS Pose Estimation to identify the positions of human bodies and parts and presented with skeletal information. YOLO-NAS provides skeletons with different colors on body parts which offers rich information on human pose data. Specific body parts play crucial roles to violent action, as they indicate aggressive intent, for instance, raised arms can indicate pushing or grabbing someone. After extracting the pose information, original background was replaced with a black background. This transformation helps remove noise from the original background, and eliminates unnecessary objects in the frames.

The next phase involves calculating the motion by calculating the differences between the current and targeted frames using frame differencing. That is,

$$\text{frame_diff}_t = \text{frame}_{t+n} - \text{frame}_t \tag{1}$$

where frame_t represents the pixel values of the frame at time step t, and frame_diff_t represents the computed difference between target frame frame_{t+n}.

The extraction of skeletal and motion information is presented in pseudocode and shown in Algorithm 1. The resulting sequences of pose estimations and frame differences provide a comprehensive set of features that can be utilized for further analysis in violence detection tasks.

B. ARCHITECTURE

Our proposed architecture as illustrated in Figure 1 consists of two streams: one processes human pose sequences without background, and the other processes motion information. This dual-stream approach enables robust violence detection by harnessing complementary information from multiple inputs or modalities.

The skeleton data, extracted via pose estimation using YOLO-NAS, focuses on structural and kinematic

Algorithm 1 Video Preprocessing

Input: Continuous frames N in the video
Output: Video frames with pose estimation with black background and frame differencing between adjacent frames
for $i \leftarrow 1$ to N **do**
    Capture current Frame N_i
    Apply gamma correction to the Frame N_i
    Extract skeletal information using YOLO-NAS Pose Estimation
    Render skeletons on black background
end for
for $j \leftarrow 1$ to $N - 1$ **do**
    Calculate frame differencing between Frame N_{j+1} and Frame N_j
end for
return N_i sequences of pose estimation and N_j sequences of frame differencing

information such as 2D joint positions and their spatiotemporal relationships. This stream captures the pose topology and body kinematics (e.g., limb orientations, joint angles) while abstracting away visual details like clothing, background, or appearance. In contrast, the motion data stream, derived from frame differencing, encodes contextual dynamics and motion cues at the pixel level, including movements and interactions between individuals and object or environment, which is not represented in the skeleton features. While both streams are designed to model temporal information via ConvLSTM layers, their resulting feature maps differ that skeleton features are sparse and geometry-focused, producing compact representations that are robust to visual clutter, whereas motion features are dense and holistic, capturing subtle visual motions but more susceptible to noise.

The human pose sequence was generated from the original RGB frames using a Pose Estimation model, YOLO-NAS. The frame difference sequence was created by subtracting consecutive frames, highlighting motion and changes in pixel intensity. The sequences were then processed independently through separate pipelines. Each pipeline consists of ConvLSTM and batch normalization layers that extract features from the input data.

Each pipeline is applied different activation functions using Tanh and ReLU, and multiplied with different scalar weights before fused with Sum fusion operation. The fusion mechanism is followed by ConvLSTM layer to maintain both spatial and temporal information. The ConvLSTM layer outputs a compact spatio-temporal feature maps and followed by simple convolutional layer for efficiency for feature learning. Before the classification layers, the global pooling layer reduces dimensions of the feature maps from the convolutional layer. Finally, the fused output is passed through a fully connected layer for classification, which predicts the probability of violence.

C. FUSION STRATEGY

Fusion strategies play a crucial role in effectively combining information from multiple sources or modalities to enhance model performance. Common fusion techniques use fusion operation functions such as concatenation, summation, averaging, and multiplication, each offering unique advantages.

Fusion in violence detection can be significantly enhanced by employing different activation functions, such as tanh and ReLU, within various pipeline architectures. The tanh function, outputs values between -1 and 1, which provides both positive and negative changes in pixel intensity. By capturing both positive and negative deviations, it can be helpful in identifying subtle changes. This is particularly beneficial when combining features from different sources, such as motion data and skeletal information, as it reduces the risk of one feature dominating due to its scale. On the other hand, the ReLU function, which outputs zero for negative inputs and the input itself for positive values. This feature captures significant changes in pixel values, while filtering out negative values and emphasizing only positive changes.

ReLU function preserves positive values without changing its values, while Tanh function captures both positive and negative values within the range of -1 to 1. By applying these two functions on branches, the intensity of the output values can be controlled. ReLU can be used to capture high-intensity features and eliminate noise, whereas Tanh can be used to capture small changes and detect subtle variations in intensity.

Another approach applied in this report is weighted outputs fusion in violence detection involves combining information from skeleton-based inputs and frame differencing inputs, by multiplying with different scalar weights to each branch output before fusion based on their relevance and contribution to the detection task. By assigning a higher weight to one of the branches, the information of that branch is prioritized in the fusion process.

By designing a pipeline that strategically incorporates both activation functions, the model can leverage the strengths of each modality such as skeleton data and frame differencing while adjusting the weights assigned to each based on their performance. This approach allows for a more adaptive focus on the most informative features, leading to improved accuracy in violence detection.

IV. EXPERIMENTS AND RESULTS

A. DATASETS

Over several datasets have been published for violence detection, the RWF-2000 dataset dataset [12] is particularly notable as it offers the largest collection of fixed-length videos. Additionally, videos in the RWF-2000 dataset [12] were captured from surveillance cameras that align with our objective of using it as a dataset for experiments. The RWF-2000 dataset is structured with predefined training and validation splits that contain 2000 videos with 20% for validation. Importantly, videos in the RWF-2000 dataset are categorized into two classes: violence and non-violence.

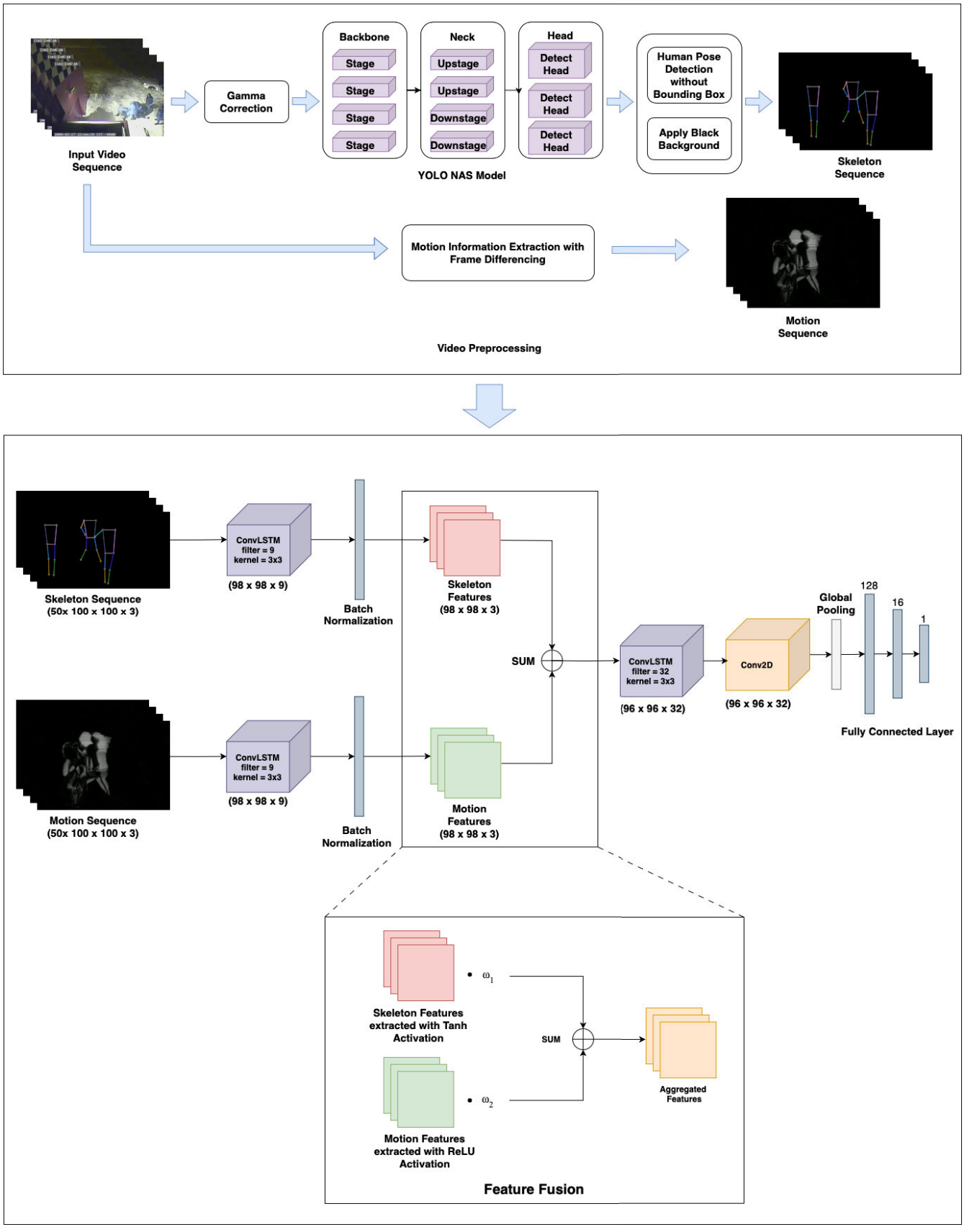


FIGURE 1. Proposed framework and architecture.

Furthermore, the proposed model is evaluated on other datasets to prove its generalization. Other datasets that are used in the experiments are Hockey [13], Movies [13], Crowd [14], RLVS [15] and SCFD [16]. These datasets were selected specifically for their relevance to violence detection. These datasets were selected specifically for their relevance to violence detection, and like the RWF-2000 dataset [12], they also consist of two classes: violence and non-violence. This consistent classification across all datasets allowed for a comprehensive evaluation of the model's performance in detecting violent behavior.

B. IMPLEMENTATION

The implementation of the model involved using 50 uniformly selected frames per video, with each frame captured at a frame rate of 30 frames per second (FPS). To standardize the input data and the efficiency of model training, each frame was resized to a resolution of 100 × 100 pixels. For training, 10 videos per batch were applied for 30 epochs to iteratively refine the parameters and improve the performance in detecting violent actions.

For the training configuration of the model, the Adam optimizer [17], which is known for its efficiency and effectiveness was used. The learning rate was set to 0.001, which balances the convergence speed and stability during training. The training process was conducted on a single NVIDIA A100 GPU, which provided the necessary computational power to efficiently handle the processing of the video data and training of the deep learning mode. Furthermore, experiments and quantitative results in Tables 1–6 were obtained using the RWF-2000 dataset with 10 epochs of training, which were used for ablation and pipeline analyses, whereas Table 7 presents results with 30 epochs of training on all datasets to enable fair comparison with state-of-the-art methods.

C. HUMAN POSE PIPELINE ANALYSIS

In this section, the contribution of the human pose pipeline is examined. To achieve this, the motion detection pipeline was removed from the model architecture shown in Figure 1 and the performances between the original video sequences without the preprocessing step and the skeleton sequences were assessed. Tests are done on different types of inputs for human pose pipeline, by comparing original video sequences and skeleton sequences with different detection thresholds.

D. DIFFERENT INPUTS ON HUMAN POSE PIPELINE ANALYSIS

Table 1 shows the performance of the architecture between original sequences without preprocessing and skeleton sequences using the RWF-2000 dataset. Initially, the violence detection performs reasonably well with original videos, achieving a training accuracy of 70.94% and a validation accuracy of 77.00%. When pose information through skeletons was incorporated, the performance improved, with training accuracy rising to 76.00% and validation accuracy to 78.45%, even in the presence of background noise. Ultimately, by eliminating the original background and replacing with a black background and retaining only the essential skeleton data, there was a significant increase in model accuracy, with training accuracy reaching 91.00% and validation accuracy at 86.75%. This reinforces our hypothesis that focusing on human bodies and their interactions is sufficient for detecting violence, and that reducing noise enhances the model's performance.

TABLE 1. Violence detection accuracies on RWF-2000 dataset using original sequences and skeletal sequences inputs.

Input	Training Accuracy	Validation Accuracy
RGB Frames	0.7094	0.7700
Skeletons with background	0.7600	0.7845
Skeletons without background	0.9100	0.8675

TABLE 2. Violence detection accuracies on RWF-2000 dataset with different pose detection confidence thresholds.

Confidence Threshold	Training Accuracy	Validation Accuracy
0.4	0.8462	0.8181
0.2	0.8388	0.8225
0.0	0.9100	0.8675

E. VIOLENCE DETECTION RESULTS WITH DIFFERENT HUMAN POSE DETECTION THRESHOLDS

As only human pose detection input is solely relied upon, there is a risk of losing information if skeletons are not accurately detected. To maximize the performance of the pose estimation model, YOLO-NAS pose detection using different confidence thresholds was also evaluated on RWF-2000 dataset. The results are shown in Table 2, indicating the impact of varying confidence thresholds on the training and validation accuracy of the pose estimation model for violence detection.

Overall, the violence detection performance on pose detection with a threshold set to 0 performs best with the training accuracy 91.00%, and the validation accuracy 86.75%, compared with others. This suggests that by allowing the model to accept all detected poses, it can learn more from the training data, resulting in higher accuracy.

F. MOTION PIPELINE ANALYSIS

After analyzing different alternatives for the RGB pipeline, the motion pipeline is then evaluated on RWF-2000. Again, to understand its potential as the sole input, different motion inputs of different frames are analyzed with the similar architecture of the human pose pipeline analysis. In this context, frame differences (FD) are analyzed using multiple settings, such as 1FD, 2FD, and so on, where each setting represents the difference between frame t and frame $t + n$. Here, nFD denotes the frame difference computed between frame t and frame $t + n$ (e.g., 1FD $= I_{t+1} - I_t$,

TABLE 3. Motion pipeline analysis for violence detection on RWF-2000 dataset with different frame differences. FD denotes the frame difference computed between frame t and frame $t+n$ (e.g., 1FD $= I_{t+1} - I_t$, 2FD $= I_{t+2} - I_t$), etc.

Frame Difference (FD)	Training Accuracy	Validation Accuracy
1 FD	0.9281	0.8175
2 FD	0.9069	0.8100
3 FD	0.8938	0.8025
4 FD	0.9181	0.7900
5 FD	0.8500	0.7775

2FD $= I_{t+2} - I_t$). The results of the motion pipeline analysis, as presented in Table 3, indicate varying performance across different inputs of frame difference. Among the configurations tested, the model with a difference in 1 frame achieved the highest training accuracy of 92.81% and validation accuracy of 81.75% with a validation loss of 0.4701. From the results, the accuracies decreases when frame difference is increased, due to the loss of contextual information. This makes the 1-frame difference inputs the most effective choice for further analysis.

G. FUSION ANALYSIS

This section explores the fusion analysis of two distinct inputs from the human pose pipeline and motion pipeline to enhance the performance of the violence detection model. By combining information from different sources, the overall accuracy and robustness of the model can be improved.

For the model fusion, several fusion strategies are deployed and experimented with. Different fusion operation functions, such as summation, multiplication, averaging, and concatenation, are evaluated based on their training and validation accuracies on violence detection. Subsequently, the impact of fusion using different activation functions was investigated, specifically Tanh and ReLU. By employing these activation functions in the pipelines, the goal is to optimize the model's ability to capture dynamic changes in the data.

Moreover, the weighted output mechanism was implemented by multiplying different scalar weights to each branch output before fusion. Through this comprehensive fusion analysis, experiments are conducted to provide insights into the effectiveness of various fusion strategies and the role of activation functions in enhancing model performance.

1) FUSION OPERATION TYPES ANALYSIS

This section presents an analysis of various fusion operation types applied to the pipelines, aimed at understanding their impact on both training and validation accuracies. By evaluating different fusion operation types that are sum, multiply, average, and concatenate on the RWF-2000 dataset, the goal is to identify the most effective method for combining features from the motion pipeline, thereby optimizing the model's ability to detect violent behavior. In terms of accuracy, as compared in Table 4, the sum operation

TABLE 4. Fusion operation type analysis with violence detection on RWF-2000 dataset.

Operation Type	Training Accuracy	Validation Accuracy
Sum	0.8931	0.8823
Multiply	0.8700	0.8550
Average	0.8850	0.8725
Concatenate	0.9438	0.8625

type performed well, yielding an accuracy of 88.23% on validation. This result indicates that the model generalizes effectively to unseen data, maintaining a strong performance despite the inherent complexity of the task. The analysis of these fusion operation types highlights the importance of feature integration in enhancing the performance of violence detection models. While the concatenation method excels in training accuracy, it is essential to consider its impact on validation performance, as indicated by the slightly lower validation accuracy compared to the sum method.

2) WEIGHTED BY ACTIVATION FUNCTION

In this experiment, the effectiveness of weighted fusion of two input pipelines using Tanh and ReLU activation functions is examined. The goal is to leverage the strengths of both activation functions to improve the overall performance of the model. The Tanh activation function, known for its ability to output values between -1 and 1, was used in one pipeline, while the ReLU activation function, which outputs values between 0 and infinity, was used in the other pipeline. The outputs of the two pipelines were then combined using a weighted sum, where the weights were adjusted to emphasize one input over the other.

In Table 5, the performance of the model using Tanh and ReLU activation functions on different pipelines are evaluated. Using the tanh activation function on frame differencing in violence detection can yield better results due to its ability to effectively model the changes between consecutive frames. The tanh function outputs values in a range from -1 to 1, which allows it to capture both positive and negative changes in pixel intensity, enabling the model to detect subtle variations in motion that may indicate violent behavior. From the Table 5, the pose pipeline using the Tanh activation function and a motion pipeline utilizing ReLU, achieved a training accuracy of 91.31% and a validation accuracy of 86.00%. The second configuration, which features a pose pipeline using ReLU and a motion pipeline employing Tanh, recorded a training accuracy of 0.9069 and a validation accuracy of 0.8775. Although the training accuracy is slightly lower than that of the first configuration, the validation accuracy is higher. This suggests that the combination of ReLU in the pose pipeline and Tanh in the motion pipeline may be facilitating better generalization to the data. This emphasizes the significance of motion and transitions in the video data.

TABLE 5. Performance comparison of different activation functions across pipelines for violence detection on RWF-2000 dataset.

Configuration	Training Accuracy	Validation Accuracy
Pose Pipeline (Tanh) and Motion Pipeline (ReLU)	0.9131	0.8600
Pose Pipeline (ReLU) and Motion Pipeline (Tanh)	0.9069	0.8775

TABLE 6. Performance of the model using weighted outputs with Sum fusion across different pipelines on RWF-2000 dataset.

Configuration	Training Accuracy	Validation Accuracy
Pose Pipeline ($\omega_1 = 1$) and Motion Pipeline ($\omega_2 = 2$)	0.8931	0.8950
Pose Pipeline ($\omega_1 = 2$) and Motion Pipeline ($\omega_2 = 1$)	0.9169	0.8850

3) WEIGHTED OUTPUT FUSION

In this section, the weighted output mechanism by multiplying with different scalar weights for each branch output before fusion is investigated. This mechanism is presented using the following equation:

$$O_{\text{Merged}} = O_{\text{Pose}} \cdot \omega_1 \ \oplus \ O_{\text{Motion}} \cdot \omega_2 \tag{2}$$

where O_{Merged} represents the combined output using fusion operation type from pose pipeline O_{Pose} and motion pipeline O_{Motion}, multiplied with scalar weights ω_1 and ω_2 respectively. For the performance evaluation, the weight ω_1 is assigned to the Pose Pipeline, and the weight is assigned to the Motion Pipeline with different values. To see the performance between different weights, the values 1 and 2 were chosen, and the performance was evaluated on violence detection. By assigning a weight of 1 to a pipeline, the model ensures that the contribution of that pipeline features is present but not overly influential. The higher weight of 2 for a pipeline allows the model to be more sensitive to the features. The distinct weight settings 1 and 2 enable to observe how changes in the weights affect the model's performance. The violence detection results of different settings are displayed in Table 6.

In the first configuration, where the pose pipeline is assigned a weight of $\omega_1 = 1$ and the motion pipeline a weight of $\omega_2 = 2$, the model achieved a training accuracy of 89.31% and a validation accuracy of 89.50%. The second configuration, which assigns a higher weight to the Pose Pipeline with $\omega_1 = 2$ and a lower weight to the Motion Pipeline with $\omega_2 = 1$, recorded a training accuracy of 91.69% and a validation accuracy of 88.50%. Although the second configuration has a higher training accuracy than that of the first, the first configuration shows higher validation accuracy with better generalization on both training and validation.

This finding highlights the role of scalar weights selection in the performance of the model. The first configuration with the weight skewed on motion pipeline shows more balanced results.

H. CROSS DATASETS ANALYSIS

To demonstrate the overall performance of the proposed model and combined fusion strategy, 5-fold cross validation is carried out on Hockey, Movie, Crowd, SCFD, RLVS, and RWF-2000 datasets. The datasets present unique challenges and characteristics in detecting violent behavior in diverse contexts. Across all datasets, 80% split of the data was used for training and 20% for the testing. The applied configuration of combined fusion strategy on the model is presented in the equation as:

$$O_{\text{Merged}} = \text{ReLU}(O_{\text{Pose}}) \cdot \omega_1 \ \oplus \ \text{Tanh}(O_{\text{Motion}}) \cdot \omega_2 \tag{3}$$

where O_{Merged} is achieved by Sum fusion operation on O_{Pose} and O_{Motion}, applied with $\omega_1 = 1$ and $\omega_2 = 2$ respectively.

The results from the 5-Fold Cross Validation across various datasets is illustrated in Table 7. Results are reported after 30 epochs of training on all datasets to ensure a fair comparison, which differs from the 10-epoch setting used in Tables 1–6. Overall, the average accuracies across all folds reveal a high level of performance for the model. The Hockey dataset achieved an average testing accuracy of 98.50%, while the Movie dataset attained 99.50%. The Crowd dataset averaged a testing accuracy of 95.40% and 91.33% on SCFD dataset. The average accuracy of RLVS is 96.80% while the RWF-2000 dataset achieved an accuracy of 95.05%.

Although the Hockey and Movie datasets show exceptional performances, there are some variations in other datasets. In the Crowd dataset results, the first four folds demonstrate promising performance, with accuracies consistently above 95%. However, the last fold presents a stark contrast, with accuracy of 82.00%. The lower accuracies indicates that the model struggled to learn from the data, and the selected data in this fold may contain more challenging or ambiguous samples, leading to difficulties in classification.

From the performance of the model on the RWF dataset, the training accuracies range from 91.69% to 96.49%, suggesting that the model is effectively capturing the underlying patterns in the training data. The validation are similarly promising, with values ranging from 91.50% to 97.00%. From these results, the model is able to generalize effectively to the data.

I. INFERENCE AND QUALITATIVE ANALYSIS

The performance of the proposed model for violence detection was evaluated using various metrics across five-fold cross-validation, yielding promising results shown in Table 8. The average recall was recorded at 96.03%, indicating the model's strong ability to correctly identify instances of violence. In terms of specificity, the model achieved an average score of 96.36% and only 1 result from five-fold is less than 95%, demonstrating its capability to accurately identify non-violent instances. This would be beneficial in real-world cases, such as when efficient security services are required and reducing unnecessary interventions. Four out of five precision scores are above 95% with average 96.30% indicating that detection of violent cases is also

TABLE 7. Performance metrics of the proposed model trained with 30 epochs across different datasets (Hockey, Movie, Crowd, SCFD, RLVS, and RWF-2000) evaluated using five-fold cross-validation. The table presents the accuracy scores for each fold along with the overall average accuracy for each dataset.

	Hockey	Movie	Crowd	SCFD	RLVS	RWF-2000
1Fold	0.9900	1.0000	1.0000	0.9167	0.9750	0.9150
2Fold	0.9900	1.0000	1.0000	0.8667	0.9475	0.9600
3Fold	1.0000	0.9750	1.0000	0.9333	0.9650	0.9700
4Fold	0.9950	1.0000	0.9500	0.9500	0.9725	0.9625
5Fold	0.9500	1.0000	0.8200	0.9000	0.9800	0.9450
Average	0.9850	0.9950	0.9540	0.9133	0.9680	0.9505

TABLE 8. Detection and inference analysis on RWF-2000 dataset.

Metric	1-Fold	2-Fold	3-Fold	4-Fold	5-Fold	Overall Average
Recall	0.9492	0.9466	0.9896	0.9948	0.9213	0.9603
Specificity	0.9360	0.9742	0.9565	0.9567	0.9946	0.9636
Precision	0.9350	0.9750	0.9550	0.9550	0.9950	0.9630
F1-score	0.9420	0.9606	0.9720	0.9745	0.9567	0.9612
Accuracy	0.9425	0.9600	0.9725	0.9750	0.9550	0.9610
Average Inference Time (s)	0.3438	0.3259	0.3269	0.3334	0.3324	0.3325
Average Memory Used (MB)	3.5373	5.8594	3.7917	3.6276	3.4762	4.0584

high and effective. The F1-score, which combines both precision and recall, averaged at 96.12% with 4 out of 5-fold evaluation achieve about 96%, highlighting the model's overall effectiveness in violence detection.

Regarding inference time, the model demonstrated an average inference time of approximately 0.3325 seconds per instance not including video preprocessing, that the model can process and predict quickly. This rapid processing capability allows the model to make predictions quickly, which is critical for applications requiring immediate responses, such as surveillance systems in public spaces or security monitoring in sensitive environments.

In addition to inference time, the model's memory usage was also evaluated. The average memory consumption across the five folds was recorded at 4.06 MB, with individual fold memory usage ranging from 3.48 MB to 5.86 MB. This relatively low memory footprint indicates that the model can be deployed on devices with limited resources, making it suitable for real-time applications in various settings.

J. COMPARISON WITH OTHER MODELS

Table 9 compares the performances of various models for violence detection, evaluated based on the validation accuracy. The models have improved over time, with the most recent ones achieving the highest accuracies. Our model achieves an accuracy of 95.05% on RWF-2000 dataset, which is comparable to the best performing models like CUE-Net [31]. This highlights the effectiveness of our model in violence detection.

Moreover, the proposed model demonstrates robust performance across multiple benchmark datasets, outperforming or matching the performance of state-of-the-art methods while maintaining significantly fewer parameters (118,755). As shown in Table 9, our approach achieves exceptional results on the Hockey dataset (98.50%), Movie dataset (99.50%), and Crowd dataset (95.40%), performing competitively with the best models in the literature. Particularly notable is our model's strong performance on the more recent and challenging RLVS dataset (96.80%), where it achieves competitive performance while utilizing substantially fewer computational resources. The model also maintains solid performance on the SCFD dataset (91.33%), demonstrating its adaptability to different datasets. Most importantly, our model achieves 95.05% accuracy on the RWF-2000 dataset, which represents one of the highest reported accuracies while maintaining parameter efficiency. The architecture is designed to enable low-latency inference and efficient deployment on resource-constrained edge devices, making it particularly suitable for real-world applications such as surveillance systems, where larger models may be impractical. Furthermore, a balance between accuracy, model complexity, and practical applicability is demonstrated, which is crucial for real-world violence detection systems.

This multi-dataset evaluation confirms that the fusion strategy maintains consistent performance across diverse violence contexts, addressing concerns about the generalizability of violence detection approaches. It is important to note that different models utilize different techniques and architectures, and their performance can vary depending on factors like dataset size and the specific definition of violence.

K. DISCUSSION

The proposed architecture shows promising results in violence detection, exceeding the performance of several existing methods on the RWF-2000 dataset. The dual-stream design, incorporating both human pose information and frame differences, proves effective in capturing a more comprehensive understanding of violent actions. Analysis

TABLE 9. Accuracy comparison with state of the art in different datasets. '-' indicates that the number was not published.

Ref	Year	Model	Datasets						Parameters
			Hockey	Movie	Crowd	RLVS	SCFD	RWF-2000	
[18]	2020	SPIL	0.9680	0.9850	0.9450	—	—	0.8930	507 154
[12]	2021	FGN + 3D CNN	0.9800	1.0000	0.8800	—	—	0.8730	—
[10]	2021	SepConvLSTM + MobileNet	0.9950	1.0000	—	—	—	0.8975	333 057
[19]	2021	2D CNN + LSTM	0.9960	1.0000	0.9800	—	—	0.8800	1.3000 M
[20]	2022	2D-3D CNN	0.9471	1.0000	0.9141	—	—	0.8198	352 000
[21]	2022	Fast-RCNN	0.9730	—	—	0.9288	—	0.8870	—
[22]	2022	MobileNetV2 + LSTM	0.9610	0.9950	—	—	—	0.8200	3.4572 M
[23]	2022	X3D	—	—	0.9800	—	0.8870	0.9400	—
[24]	2022	ConvLSTM + GRU	0.9850	—	—	—	0.7590	0.8820	—
[7]	2022	Two-streamed ConvLSTM	0.9450	0.9850	0.9430	—	—	0.9025	62 583
[25]	2023	3D CNN	0.9750	1.0000	0.9950	0.9670	—	0.9100	29.7672 M
[26]	2024	X3D	—	—	—	—	0.9130	0.8720	—
[27]	2024	lightweight MobileViT	0.9850	1.0000	—	—	—	0.8630	—
[28]	2024	Pretrained CNN models + LSTM	—	—	—	0.9630	—	0.9270	—
[29]	2024	GCNVertexEdge GNN	0.7139	0.7730	—	—	—	0.6223	475
[30]	2024	3D ResNet	—	—	—	—	—	0.9050	66 M
[31]	2024	UniformerV2 + Modified Efficient Additive Attention	—	—	—	0.9950	—	0.9400	354 M
[32]	2024	Improved EfficientNet-B0 + Bi-directional motion attention	0.9850	1.0000	—	—	0.9167	0.9025	4.48 M
		Ours	0.9850	0.9950	0.9540	0.9680	0.9133	0.9505	118 755

of different types of fusion operation functions, weighted strategies with weighted output on branches, and activation functions highlights the importance of strategically combining the two modalities to maximize performance. The analysis on cross-datasets indicates the model's strength in structured environments but reveals challenges in handling complex and ambiguous scenarios, as in the SCFD, Crowd, and RWF-2000 datasets.

V. CONCLUSION

With the importance of surveillance and security in various sectors, including public safety, law enforcement, and private security, the growing need for advanced technologies that can automatically analyze video feeds to detect and respond to potential threats has become significant. The ability to accurately identify violent behavior in video sequences is important to prevent violent incidents.

This research presents a novel dual-stream architecture for violence detection in video sequences. Using human pose and motion information, the model captures a comprehensive representation of violent behavior. The proposed architecture demonstrates superior performance on the RWF-2000 dataset, exceeding the accuracy of existing methods with competitive performance on other datasets. Analysis of different fusion strategies provides valuable insights on the fusion configurations to effectively exploit information from different modalities to optimize model performance.

While the findings indicate the effectiveness of the fusion strategy on the evaluated datasets, it should be acknowledged that claims regarding the fusion strategy in this study are context-specific to the RWF-2000 dataset. Further validation on additional diverse datasets would be valuable and represents an important direction for future work to strengthen generalizability claims. Additionally, limitations in the approach should be noted, such as the absence of appearance information, which is critical for distinguishing violent acts from benign interactions such as playful gestures.

To further enhance feature integration for violence detection, future work will explore more advanced fusion methods. Specifically, sophisticated attention mechanisms, such as multi-head attention or transformer-based fusion, will be investigated to better capture complex spatio-temporal dependencies in video data. Future research may also consider incorporating additional modalities, such as audio data, to provide a more comprehensive understanding of violent incidents, further improving detection accuracy and contextual awareness.

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SAI THU YA AUNG received the B.C.Tech. degree (Hons.) in computer technology from the University of Computer Studies, Yangon, Myanmar, and the M.Sc. degree in computer science from the Faculty of ICT, Mahidol University, Thailand, where he is currently pursuing the Ph.D. degree in computer science. His research interests include developing and applying machine learning, deep learning, and image and video analysis to solve problems in a variety of domains, including smart surveillance, smart agriculture, and bioinformatics.



WORAPAN KUSAKUNNIRAN (Member, IEEE) received the B.Eng. degree in computer engineering from the University of New South Wales (UNSW), Sydney, NSW, Australia, in 2008, and the Ph.D. degree in computer science and engineering from UNSW, in cooperation with the Neville Roach Laboratory, National ICT Australia, Sydney, in 2013. He is currently an Associate Professor with the Faculty of Information and Communication Technology, Mahidol University, Nakhon Pathom, Thailand. He is the author of several papers in top international conferences and journals. He was a program committee member of many international conferences and workshops. He was also a reviewer of several international conferences and journals, such as ICPR, ICIP, PR, TIP, and TIFS. His research interests include biometrics, pattern recognition, medical image processing, computer vision, multimedia, and machine learning. He was a recipient of the ICPR Best Biometric Student Paper Award, in 2010, and also a winner of several national and international innovation contests.



YEW KWANG HOOI received the Ph.D. degree from Universiti Teknologi PETRONAS, Malaysia. He is currently a Senior Lecturer with the Department of Computer and Information Sciences, Universiti Teknologi PETRONAS. His research interests include semantic web, knowledge representation, and formal language.